Max Conway

max.conway@colorado.edu
https://maxconwa.github.io/

PROFESSIONAL EXPERIENCE

University of Colorado Boulder — **Boulder, CO**
Teaching Assistant — *Aug 2024 – Dec 2024*
Research Assistant — *Aug 2024 – Now*

- Planning in world models within an option space instead of a classic action space. Using SLURM and High Performance Compute Clusters to use reinforcement learning for learning control policies and world models.
- Game Theoretic Humanoid Control Learns a pair of lower and upper body policies that must coordinate while achieving their own objectives using reinforcement learning and Isaac Lab..
- Led a 17-person team developing GOLEM, a modular humanoid autonomy stack enabling high-precision industrial tasks including EV battery disassembly at millimeter-level accuracy (golem-humanoid.github.io)
- Designed and deployed a robotic arm platform with linear actuator and Vision Language Model (VLM) for autonomous EV battery disassembly, owning motion planning, perception, and mechanical design (CASE 2026)

University of Denver — **Denver, CO**
Adjunct Professor — *June 2024 – Aug 2024*
Research Assistant — *Sept 2022 – Aug 2024*
Teaching Assistant — *Nov 2023 – June 2024*

- Developed and taught a course on applied machine learning that taught students machine learning theory, including regressions, gradient descent, back propagation, multilayer perceptrons, and transformers. By the end of the class students had developed image classification and reinforcement learning projects in PyTorch.
- Teaching assistant for four upper level computer science classes: Machine Learning, Software for AI Robotics, Embedded Systems, and Advanced Algorithms. All of these classes I previously excelled in and loved.
- Researched multimodal sensor fusion across thermal, depth and RGB cameras for human activity recognition at the ARISE robotics lab (arise-lab.com). Supervised by Dr. Christopher Reardon. Paper that was accepted to IROS 2024

DEVCOM Army Research Laboratory — **Adelphi, MD**
Masters Research Intern — *June 2023 – Sept 2023*

- Worked on fundamental research problems in the intersection of computer vision and knowledge representations. My work at ARL was focused on perception for autonomous robotics. This work continued from a summer internship into a collaboration between ARL and the University of Denver.

Amazon, Inc. — **Remote from Denver, CO**
Software Engineer Intern — *June 2022 – Sept 2022*

- Designed, coded and delivered improved notification experience for multimedia Alexa devices. Wrote design document, learned new technology stack and worked with UX team to complete internship project ahead of schedule. Received offer to return as full time software engineer.

EDUCATION

University of Colorado Boulder, — **Boulder, CO**
Candidate for PhD in Computer Science — *Expected to finish in 2028*
GPA: 4.0

- **Academics:** Dean's & Departmental Excellence Fellowship
- **Relevant Coursework:** Advanced Computer Vision, Transformers in Robotics, Decision Making under Uncertainty, Reinforcement Learning, Vision Language Models for Robotics

University of Denver, Ritchie School of Engineering and Computer Science — **Denver, CO**
M.S. & B.S. in Computer Science. Minors: Physics & Math — *Masters complete December 2023*
M.S. GPA:4.0, B.S. GPA: 3.85

- **Academics:** Dean's List, Hornbeck Scholar.
- **Relevant Coursework:** Machine Learning, Software for AI Robotics, Programming Languages

PUBLICATIONS

- Max Conway, Brian Reily, Christopher Reardon: **Learned Sensor Fusion For Robust Human Activity Recognition in Challenging Environments** (Accepted to IROS 2024)
- Andrew Hannum, Max Conway, Andre Harrison, Mario Lopez: **Data-Driven Hierarchical Open Set Recognition** (Accepted to ICRA@40)
- Zachary Allen*, Max Conway*, Lyle Antieau, Allen Ponraj, Nikolaus Correll: **Robotic Agentic Platform for Intelligent Electric Vehicle Disassembly** (Accepted to CASE 2026)
- Max Conway, et al: **GOLEM: Modular Humanoid Autonomy Towards Electric Vehicle Battery Disassembly** (Under Review at IEEE-RAS Humanoids 2026)